CLASS 10 MATHEMATICS :: SLIP TEST – FA-3

Maximum Marks: 20 • Time: 45 Minutes

Name: _____ **Roll No:** _____ **Sec:** _____ **Marks:** _____

SECTION I: Short Answer Questions — 3 × 2M = 6 Marks

1. Find the distance between the points A(−3, 4) and B(5, −2) using the distance formula. **[2 M]**

2. In ΔABC, DE || BC. If AD = 2.4 cm, DB = 3.6 cm, and AE = 2.0 cm, calculate the length of EC. **[2 M]**

3. A point P is at a distance of 13 cm from the centre of a circle of radius 5 cm. Find the length of the tangent drawn from P to the circle. **[2 M]**

SECTION II: Long Answer Questions — 2 × 4M = 8 Marks

4. Find the area of the triangle formed by the points A(1, 1), B(5, 1), and C(5, 4) using: **[4 M]**
(i) The basic formula: (1/2) × base × height.
(ii) The coordinate geometry formula: $(1/2)\,|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$.
Verify that both methods yield identical results.

5. State and prove the Basic Proportionality Theorem (Thales Theorem). **[4 M]**

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Multiple Choice Section

SECTION III: Multiple Choice Questions **6 × 1M = 6 Marks**

6. The midpoint of the line segment joining P(−4, 6) and Q(8, −2) is:

(A) (2, 2) (B) (4, 4) (C) (−2, 4) (D) (6, 2)

7. If the points (1, 2), (0, 0), and (a, b) are collinear, then:

(A) a = b (B) a = 2b (C) 2a = b (D) a + b = 0

8. The perimeters of two similar triangles ΔABC and ΔPQR are 36 cm and 24 cm. If PQ = 10 cm, then AB is:

(A) 12 cm (B) 15 cm (C) 18 cm (D) 20 cm

9. If tangents PA and PB from point P to a circle with centre O are inclined at 70°, then ∠POA is:

(A) 55° (B) 65° (C) 70° (D) 110°

10. The length of an arc of a circle of radius 21 cm which subtends an angle of 60° at the centre is:

(A) 11 cm (B) 22 cm (C) 33 cm (D) 44 cm

11. A tangent to a circle intersects the circle in:

(A) Exactly one point (B) Two points (C) No point (D) Infinitely many points

CLASS 10 MATHEMATICS :: SOLUTIONS – FA-3

Section I & Section II Solutions

SECTION I: Short Answer Solutions — 6 Marks

Solution 1 (Distance Formula) — 2 M

$d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$

$= \sqrt{[(5 - (-3))^2 + (-2 - 4)^2]} = \sqrt{[(8)^2 + (-6)^2]}$

$= \sqrt{(64 + 36)} = \sqrt{100} = 10$ units

Distance AB = 10 units

Solution 2 (Thales Theorem Ratio) — 2 M

Since DE || BC in ΔABC: AD / DB = AE / EC

2.4 / 3.6 = 2.0 / EC ➔ 2/3 = 2.0 / EC

2 × EC = 6.0 ➔ EC = 3.0 cm

EC = 3 cm

Solution 3 (Tangent Length) — 2 M

Radius OP ⊥ Tangent PT ➔ ΔOPT is a right-angled triangle.

$PT = \sqrt{(OP^2 - OT^2)} = \sqrt{(13^2 - 5^2)}$

$= \sqrt{(169 - 25)} = \sqrt{144} = 12$ cm

Length of tangent = 12 cm

SECTION II: Long Answer Solutions — 8 Marks

Solution 4 (Dual Triangle Area) — 4 M

A(1, 1), B(5, 1), C(5, 4)

Method 1: Base AB = 5 − 1 = 4 units, Height BC = 4 − 1 = 3 units

Area = (1/2) × base × height = (1/2) × 4 × 3 = 6 sq. units

Method 2: Area = $(1/2) |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$

$= (1/2) |1(1 - 4) + 5(4 - 1) + 5(1 - 1)| = (1/2) |1(-3) + 5(3) + 5(0)|$

$= (1/2) |-3 + 15 + 0| = (1/2) |12| = 6$ sq. units

Method 1 = Method 2 = 6 sq. units (Verified)

Solution 5 (BPT Proof) — 4 M

Statement: If a line is drawn parallel to one side of a triangle intersecting other two sides, it divides the two sides in the same ratio.

Given: ΔABC, DE || BC intersecting AB at D and AC at E.

Construction: Join BE and CD; draw DM ⊥ AC and EN ⊥ AB.

Area(ΔADE) = (1/2) × AD × EN, Area(ΔBDE) = (1/2) × DB × EN ➔ Area(ΔADE)/Area(ΔBDE) = AD/DB ... (1)

Area(ΔADE) = (1/2) × AE × DM, Area(ΔCDE) = (1/2) × EC × DM ➔ Area(ΔADE)/Area(ΔCDE) = AE/EC ... (2)

Since ΔBDE and ΔCDE share same base DE and are between same parallels DE and BC, Area(ΔBDE) = Area(ΔCDE).

From (1) and (2): AD / DB = AE / EC (Hence proved)

CLASS 10 MATHEMATICS :: SOLUTIONS – FA-3

Section III Multiple Choice Solutions

SECTION III: Multiple Choice Solutions — **6 Marks**

Solution 6 — 1 M

M = $((x_1+x_2)/2, (y_1+y_2)/2)$

= $((-4+8)/2, (6+(-2))/2) = (4/2, 4/2) = (2, 2)$

Answer: (A) (2, 2)

Solution 7 — 1 M

Collinear condition (Area = 0):

$1(0 - b) + 0(b - 2) + a(2 - 0) = 0$

$-b + 2a = 0 \Rightarrow 2a = b$

Answer: (C) 2a = b

Solution 8 — 1 M

Perimeter(ΔABC)/Perimeter(ΔPQR) = AB/PQ

36/24 = AB/10 ➔ 3/2 = AB/10 ➔ AB = 15 cm

Answer: (B) 15 cm

Solution 9 — 1 M

∠AOB = 180° − 70° = 110°

∠POA = ∠AOB / 2 = 110° / 2 = 55°

Answer: (A) 55°

Solution 10 — 1 M

$l = (\theta/360°) \times 2\pi r$

$= (60°/360°) \times 2 \times (22/7) \times 21$

$= (1/6) \times 2 \times 22 \times 3 = 22$ cm

Answer: (B) 22 cm

Solution 11 — 1 M

By definition, a tangent intersects a circle at exactly one point.

Answer: (A) Exactly one point